

RAMKUMAR S.

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1 Career objective

The objective is to work in the field of Computational Physics, experienced in Computational Fluid Dynamics and Scientific Machine Learning.

2 Academics

course	Institution	Board or University	Year of completion	Marks
10 th	Aditya Vidhyashram	CBSE	2010	8.4 CGPA
12 th	Aditya Vidhyashram	State Board	2012	96.4%
B.E. Aeronautical	PARK Engineering College	Anna University	2016	7.8 CGPA
M.Tech Aerospace	Indian Institute of Space Science and Technology	Autonomous	2024	7.5 CGPA

3 Work Experience

Designation	Organization/Company	Duration	Experience
Freelancer in CFD simulations	self	Apr-2016 to Apr-2018	2 years
Junior Research Fellow	CFDVRI, Bangalore	May-2018 to Apr-2019	1 year
CFD Engineer	ACRi Info Tech, Bangalore	May-2019 to July-2022	3 years & 2 months

4 Technical Skills

- Software skills: OpenFOAM, ParaView, Salome, Ansys Fluent, SpaceClaim
- Platform skills: Linux (Ubuntu & OpenSUSE) and Windows.
- Programming skills: Python, FORTRAN, MATLAB, C++, L^AT_EX
- Communication skills: good with English & Tamil
- Knowledge: good in the field of Computational Fluid Dynamics and Scientific Machine Learning, interested in Research and Development.

5 GATE score

Year	Registration ID	Paper	All India Rank	Score
2020	AE20S47301004	Aerospace	195	530
2021	AE21S27301042	Aerospace	178	536
2022	AE22S87300078	Aerospace	140	582

6 Top project works

1. System identification of an airplane using Physics Informed Neural Networks (PINNs)
 - System identification for the longitudinal dynamics of Hansa airplane was performed in the work.
 - A linear aerodynamic model and the dynamics governing equations in wind axis reference frame were implemented.
 - Python programming language with Tensorflow library was used in the work. Estimations were performed using both synthetic and experimental data.
 - A new PINN architecture, System-Identification PINNs (SI-PINNs) was developed, overcoming limitations faced by existing models in the literature.
 - Performance of SI-PINNs was compared with classical methods, and the advantages of SI-PINNs in handling sparse and faulty data were demonstrated.
 - SI-PINN is proposed as an alternative to classical methods in the aircraft system identification, and the experimentations can be geared towards sparse but accurate data instead of huge and faulty.
2. Development of custom OpenFOAM solver for the wave equation
 - A custom application named *waveFoam* was developed using OpenFOAM C++ to solve the transient wave equation.
 - The solver was developed to handle a maximum of upto 4 dimensional solution field (3 spatial and 1 temporal) and embodied with all features that a basic OpenFOAM application has.
 - A custom foam dictionary was made to feed the wave speed property needed by the solver that can access it during execution.
 - A new field file *h* was created that records the amplitude of waves in the domain.
 - The solution files were saved in FOAM format to post-process directly using ParaView.
 - The code was submitted as a tutorial for the blooming programmers in OpenFOAM community. <https://github.com/UnnamedMoose/BasicOpenFOAMProgrammingTutorials>
3. Development of incompressible single-phase massless particle tracking application in OpenFOAM
 - There are few particle tracking applications in OpenFOAM v1806 but none of them are massless, hence this application was developed.
 - The application tracks particles by displacing them with integration of fluid velocity over a given timestep. The particles are represented as points in 3D domain.
 - A new foam dictionary, *particleTracksDict* was developed that stores the initial position of particles, which are defined using either a list of coordinates or by defining the name of a boundary.
 - Back or forward tracing will be determined by the application based on the specified inlet or outlet boundary.
 - The application writes residence time and distance travelled by each particle to a post-processing file and the path of each particle is exported as vtk file for ease of visualization in ParaView.
 - The work was submitted as a tutorial to the OpenFOAM community. <https://github.com/UnnamedMoose/BasicOpenFOAMProgrammingTutorials>
4. Numerical analysis of a laminar incompressible viscous flow over flat plate and the driven cavity flow using Lattice Boltzmann Method (LBM)

- Flow simulation was carried using air at standard condition and for Reynolds numbers of 10 and 1000 for the flat plate and cavity flow, respectively.
- The Lattice Boltzmann Method (LBM), a mesoscopic approach was implemented in the work.
- The code was developed on MATLAB and Python languages.
- Validation was performed using Blasius solution for flat plate and the benchmark results from the literature for the cavity flow.
- The performance of LBM was compared to the finite difference method (FDM) for both the problems.

5. Development of a two-phase fluid flow solver for rectangular structured domains

- The solver was developed using staggered-finite volume method using FORTRAN-2008 for computation, NetCDF for data handling and Python 3.6 for post-processing.
- Immersed interface model and the Eulerian Volume of Fluid (VOF) method were implemented for the two phase flow in the work.
- Continuum surface force model was implemented to include surface tension forces in the computation.
- Pressure correction technique with modified SIMPLE algorithm was used to solve the incompressible governing equations.
- Drop at 0-gravity and the Rayleigh-Taylor instability problems were solved for the validation of the solver.

6. Numerical simulation of Poiseuille flow

- Axisymmetric Navier-Stokes equations using finite volume method using vorticity-streamfunction formulation were used in the work.
- CFD code was developed using MATLAB and Python. A simple GUI was developed using Python language.
- Validation was performed using Hagen-Poiseuille equation and comparing velocity profile with theoretical variation.

7 Github page

This page contains some of my code works <https://github.com/Ramkumar47>

8 References

- Dr. Madhukar M. Rao, Technical director, ACRi Info Tech, Bangalore. madhukar.rao@acricfd.com
- Dr. Devendra Prakash Ghate, Assistant Professor, Indian Institute of Space Science and Technology. devendra.ghate@iist.ac.in

9 Declaration

I, hereby, declare that the above given information is true to the best of my knowledge.

Date:

signature: (Ramkumar S)